



Community-acquired pneumonia in children: Clinical features and diagnosis

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INTRODUCTION

Community-acquired pneumonia (CAP) is defined by signs and symptoms of an acute infection of the pulmonary parenchyma in an individual who acquired the infection in the community, as distinguished from hospital-acquired (nosocomial) pneumonia ([table 1](#)) [1,2]. CAP is a common and potentially serious illness with considerable morbidity. Mild pneumonia is sometimes referred to as "walking pneumonia" because the symptoms do not interfere with normal activity.

The clinical features and diagnosis of CAP in children will be reviewed here. The epidemiology, pathogenesis, and treatment of pneumonia in children are discussed separately.

- (See "[Pneumonia in children: Epidemiology, pathogenesis, and etiology](#)".)
- (See "[Community-acquired pneumonia in children: Outpatient treatment](#)".)
- (See "[Pneumonia in children >28 days old: Inpatient treatment](#)".)

CLINICAL PRESENTATION

Clinical features

- **Features common to all ages** – Children with community-acquired pneumonia (CAP) typically have a combination of fever and cough and possibly other respiratory findings (eg, tachypnea, increased work of breathing), depending on the severity of the infection. Cough is triggered by the products of infection, which irritate cough receptors in the airways. In one study of children hospitalized for radiographically confirmed pneumonia, 95 percent had cough, 90 percent had fever, 75 percent had anorexia, 70 percent had dyspnea, and 55 percent had chest

wall retractions [3]. Although no single symptom or sign is pathognomonic for pneumonia in children, the longer that fever, cough, and respiratory findings are present, the greater the likelihood of pneumonia [4,5].

- **Infants and young children** – Young infants may present with difficulty feeding, restlessness, or fussiness rather than with cough and/or abnormal breath sounds [6]. Symptoms and signs of pneumonia may be subtle. Infants and young children may present with only fever and leukocytosis [5,7]. The clinical features of neonatal pneumonia are discussed separately. (See "[Neonatal pneumonia](#)", section on '[Clinical presentation](#)'.)
- **Older children and adolescents** – Older children and adolescents may experience pleuritic chest pain (pain with respiration), but this is an inconsistent finding [4]. Occasionally, the predominant manifestation in older children and adolescents is abdominal pain (because of referred pain from the lower lobes) or nuchal rigidity (because of referred pain from the upper lobes).

Bacterial versus viral etiology — The features that are traditionally considered characteristic of typical bacterial pneumonia, atypical bacterial pneumonia, and viral pneumonia ([table 2](#)) frequently overlap and cannot be used reliably to distinguish between the various etiologies [8,9]. In addition, as many as 50 percent of infections are mixed bacterial/viral infections. (See "[Pneumonia in children: Epidemiology, pathogenesis, and etiology](#)", section on '[Community-acquired pneumonia](#)'.)

Nevertheless, the diagnosis of bacterial versus atypical bacterial versus viral CAP is generally made clinically, and the following features are typically seen with each pneumonia category:

- **Bacterial** – Children with "typical" bacterial pneumonia tend to have an abrupt onset of symptoms (eg, fever, chills, cough, focal chest pain), which may be preceded by days of upper respiratory tract infection symptoms. Febrile patients usually appear ill and sometimes toxic. Respiratory distress may be moderate to severe; auscultatory findings may be few and focal, limited to the involved anatomic segment.

Pneumococcal pneumonia is the most common cause of typical bacterial pneumonia in children of all ages. Pneumonia due to *Haemophilus influenzae* type b (Hib) may still occur in areas with low Hib conjugate vaccine uptake. Typable non-type b and nontypable *H. influenzae* are uncommon causes of childhood pneumonia. Typical bacterial pneumonia is less commonly caused by *Staphylococcus aureus* and group A *Streptococcus*. Fever and cough occur in the majority of children with typical bacterial pneumonia. Other common findings include malaise/lethargy, decreased breath sounds, and crackles. Pneumococcal pneumonia in children is discussed in greater detail separately. (See "[Pneumococcal pneumonia in children](#)".)

In patients with severe illness, signs and symptoms of sepsis and localized chest pain (signifying pleural irritation) are more suggestive of a bacterial etiology because they are

rarely present in nonbacterial pneumonia [10]. Complications of pneumonia are also more suggestive of bacterial etiology ( image 1). (See '[Complications](#)' below.)

- **Atypical bacterial** – "Atypical" bacterial pneumonia usually presents abruptly with constitutional findings (eg, fever, malaise, myalgia, headache, photophobia, sore throat, conjunctivitis, rash) that gradually worsen. These are followed by a nonproductive cough that gradually worsens even as the other symptoms improve [11,12]. Wheezing is a frequent finding in atypical bacterial pneumonia [10].

Mycoplasma pneumoniae or *Chlamydia pneumoniae* are the usual causes of atypical bacterial pneumonia. *M. pneumoniae* is the most common cause of bacterial pneumonia in otherwise healthy children ≥ 5 years of age [13]. Although hoarseness may be seen in disease caused by both pathogens, it is more frequently seen with *C. pneumoniae* infection. (See "[Pneumonia caused by *Chlamydia pneumoniae* in children](#)" and "[Mycoplasma pneumoniae infection in children](#)", section on '[Clinical manifestations](#)'.)

M. pneumoniae may be associated with a variety of extrapulmonary manifestations. Dermatologic manifestations may range from a mild erythematous maculopapular rash or urticaria to *M. pneumoniae*-induced rash and mucositis (MIRM) and Stevens-Johnson syndrome. Other extrapulmonary manifestations include hemolytic anemia, polyarthrititis, pancreatitis, hepatitis, pericarditis, myocarditis, and neurologic complications [14,15]. (See "[Mycoplasma pneumoniae infection in children](#)", section on '[Clinical manifestations](#)'.)

Young infants may develop "afebrile pneumonia of infancy." Afebrile pneumonia of infancy is a syndrome generally seen between one to three months of age, but it can occur as early as two weeks or as late as four months of age. It is classically caused by *Chlamydia trachomatis*, but other agents (such as cytomegalovirus, *Mycoplasma hominis*, and *Ureaplasma urealyticum*) are also implicated. The clinical presentation is one of insidious onset of rhinorrhea and tachypnea followed by a staccato cough pattern (individual coughs separated by inspirations). Physical examination typically reveals diffuse inspiratory crackles. Conjunctivitis may be present, or there may have been a past history of conjunctivitis [16]. (See "[Chlamydia trachomatis infections in newborns and young infants](#)", section on '[Pneumonia](#)'.)

- **Viral** – The onset of viral pneumonia is gradual and associated with preceding upper respiratory tract symptoms (eg, rhinorrhea, congestion). The child does not typically appear toxic. Auscultatory findings are usually diffuse and bilateral. In one study of 98 ambulatory children with pneumonia, wheezing was more frequent in patients with viral than bacterial pneumonia (43 versus 16 percent), but other clinical features often associated with viral illness (such as rhinorrhea, myalgia, and ill contacts) were not [17].

In children younger than two years of age, >80 percent of CAP infections are caused by viruses [3].

Some viral causes of pneumonia are associated with characteristic dermatologic findings:

- Measles ( picture 1A-B) (see "Measles: Clinical manifestations, diagnosis, treatment, and prevention")
- Varicella ( picture 2) (see "Clinical features of varicella-zoster virus infection: Chickenpox")

Clinical course

- **Typical and atypical bacterial pneumonia** – Children who are appropriately treated for CAP generally show signs of improvement within 48 to 72 hours and continue to gradually improve with time. They may continue to have a residual cough for several weeks and have moderate dyspnea on exertion for two to three months.
- **Viral pneumonia** – Children with viral pneumonia generally show signs of improvement within three to seven days, with full resolution of their symptoms by one month. Rarely, the cough may last for several weeks depending on the causative agent (eg, severe acute respiratory syndrome coronavirus 2 [SARS-CoV-2]).

CLINICAL EVALUATION

Initial evaluation for all patients

Objectives — Goals of the initial evaluation include: identifying the clinical syndrome (eg, pneumonia versus bronchiolitis or asthma); determining the likely etiology (eg, bacteria, virus) ( table 2); and assessing the severity of the illness [6]. The severity of illness helps determine whether hospitalization is indicated.

History — Important aspects of the history in children with possible community-acquired pneumonia (CAP) include age of the child, details about the presenting symptoms, and a previous history of pneumonia or antibiotic use. These and other aspects of the history are outlined in the table ( table 3) [4,18]. Historical features may help determine the presumptive microbiologic etiology ( table 4), the likelihood of infection with an organism that is resistant to antibiotics, the risk of a missed foreign body or anatomical abnormality, and the severity of illness ( table 5). (See "Pneumonia in children: Epidemiology, pathogenesis, and etiology", section on 'Etiologic agents'.)

Physical examination — Important aspects of the examination include the following ( table 6):

- **General appearance** – In the young infant, assessment of general appearance includes the ability to attend to the environment, to feed, to vocalize, and to be consoled. State of awareness and cyanosis should be assessed in all children, although children may be hypoxemic without cyanosis. Most children with radiographically confirmed pneumonia appear ill [19]. Infants younger than three months of age who are toxic appearing require a

full evaluation for sepsis and other serious bacterial infections. (See ["The febrile infant \(29 to 90 days of age\): Outpatient evaluation"](#), section on 'Ill-appearing infants'.)

- **Fever** – Fever is a common manifestation of pneumonia in children [10]. However, it is nonspecific and variably present. Infants younger than three months of age who are febrile may require a full evaluation for sepsis and other serious bacterial infections. (See ["The febrile infant \(29 to 90 days of age\): Outpatient evaluation"](#), section on 'Evaluation'.)

Fever may be the only sign of occult pneumonia in young children with fever $\geq 39^{\circ}\text{C}$ (102.2°F). In one report, 26 percent of 146 children (<5 years) with fever $\geq 39^{\circ}\text{C}$, no clinical evidence of pneumonia or other localizing signs, and peripheral white blood cell count $\geq 20,000/\text{microL}$ had radiographic evidence of pneumonia [7].

Young infants ages two weeks to four months may have afebrile pneumonia caused by *C. trachomatis* or other pathogens. These children often have milder symptoms than CAP caused by other bacteria. (See ["Chlamydia trachomatis infections in newborns and young infants"](#), section on 'Pneumonia'.)

- **Respiratory distress** – Signs of respiratory distress include tachypnea, hypoxemia (peripheral arterial oxygen saturation [SpO_2] <90 percent on room air), increased work of breathing (intercostal, subcostal, or suprasternal retractions; nasal flaring; grunting; use of accessory muscles), apnea, and altered mental status [1]. Hypoxemia is a sign of severe disease and an indication for hospital admission [1,2]. The presence of grunting has been associated with severe disease and impending respiratory failure [20].

Overall, signs of respiratory distress are more specific than fever or cough for a diagnosis of pneumonia but less sensitive [4]. Children with pneumonia often have tachypnea, but tachypnea is less predictive of radiographically confirmed pneumonia than other signs of respiratory distress, especially early in the course of illness (eg, less than three days) [4,21]. On the other hand, lack of tachypnea may be helpful in excluding pneumonia. In a systematic review of studies evaluating the probability of radiographically confirmed pneumonia in children based on clinical examination findings, the probability of pneumonia was lower if the respiratory rate was ≤ 40 breaths per minute (likelihood ratio [LR] 0.41, 95% CI 0.17-0.99) [4].

Signs of respiratory distress that are more predictive of pneumonia include retractions, head bobbing, nasal flaring, and hypoxemia (defined differently in different studies, usually SpO_2 <94 to 96 percent in room air) [4,19,22,23]. However, the absence of these findings does not exclude pneumonia. In a systematic review of studies evaluating the probability of radiographically confirmed pneumonia based on clinical examination findings, the probability of pneumonia was higher if signs of increased work of breathing (eg, retractions, nasal flaring, grunting) were present (LRs 1.9-2.7, 95% CI 1.2-5.1) [4]. The absence of these findings had minimal impact on the probability of pneumonia, which was not statistically significant.

- **Lung examination** – Auscultation of all lung fields should be performed in children with suspected pneumonia. However, with the exception of wheezing, auscultatory findings have less interobserver agreement than visible findings, such as retractions or increased respiratory rate [24].

Examination findings consistent with radiographically confirmed pneumonia include [25,26]:

- Crackles
- Signs of consolidated lung parenchyma
 - Decreased breath sounds
 - Bronchial breath sounds (louder than normal and higher-pitched during expiration, with short inspiratory and long expiratory phases)
 - Egophony (E to A change)
 - Bronchophony (the distinct transmission of sounds such as the syllables of "ninety-nine")
 - Whispered pectoriloquy (transmission of whispered syllables)
 - Increased tactile fremitus (increased vibration when the patient says "ninety-nine")
 - Dullness to percussion
- Wheezing – Wheezing is more common in pneumonia caused by atypical bacteria and viruses [12,27] than typical bacteria (see '[Bacterial versus viral etiology](#)' above); also a sign of bronchiolitis or asthma
- Signs of pleural effusion – Findings suggestive of pleural effusion include chest pain with splinting, dullness to percussion, distant breath sounds, and pleural friction rub (see '[Epidemiology, clinical presentation, and evaluation of parapneumonic effusion and empyema in children](#)', section on '[Clinical presentation](#)')

Radiographs — Children with mild pneumonia who are well enough to be treated as outpatients do not require chest radiographs to confirm the diagnosis [28,29]. However, if a child has a history of a previous pneumonia episode, the child should have chest radiographs obtained to evaluate for a missed foreign body or anatomic abnormality (ie, risk factors for recurrent pneumonia).

Establishing the diagnosis — The diagnosis of pneumonia is established in a child who has fever and cough, evidence by history or on physical examination of an infectious pulmonary process ([table 3](#) and [table 6](#)), and symptoms or signs of respiratory distress. Tachypnea, nasal flaring, grunting, retractions, and hypoxia increase the likelihood of pneumonia [4]. The absence of tachypnea may be helpful in excluding pneumonia; however, the absence of other

respiratory signs does not [4]. Many children with a viral upper respiratory tract infection have cough and fever; however, they do not have signs of respiratory distress. Although no single symptom or sign is pathognomonic for pneumonia in children, the longer that fever, cough, and respiratory findings are present, the greater the likelihood of pneumonia [4,5]. Neither a radiologic nor microbiologic evaluation is necessary to confirm the diagnosis of mild CAP in children (table 5) [30].

In resource-limited countries where there is a high prevalence of pneumonia, a single positive respiratory sign increases the certainty of pneumonia [6]. The World Health Organization uses tachypnea (>60 breaths per minute in infants <2 months; >50 breaths per minute in infants 2 to 12 months; >40 breaths per minute in children 1 to 5 years; and >20 breaths per minute in children ≥5 years) as the sole criterion to define pneumonia in children with cough or difficulty breathing [31]. In one study conducted in rural Uganda, the addition of a semiquantitative point-of-care C-reactive protein (CRP) test result >40 mg/L was used as a criterion for the prescription of an antibiotic [32]. During the preintervention period, 92 percent of children with tachypnea received an antibiotic prescription compared with 71 percent during the intervention period, with no increase in adverse outcomes.

In resource-abundant countries with a lower prevalence of pneumonia, the presence of multiple respiratory signs (eg, hypoxia, grunting, nasal flaring, retractions) in addition to tachypnea is used to increase the certainty of pneumonia [4,33].

Assessing severity — An assessment of pneumonia severity is necessary to determine the appropriate setting of care. The severity of pneumonia is generally assessed by the child's degree of respiratory distress and overall clinical appearance and behavior, including an assessment of their degree of awareness and willingness to eat or drink (table 5) [1,2].

Signs and symptoms that are consistent with mild pneumonia include mild or no tachypnea, normal SpO₂ (≥90 percent on room air), and normal mental status. Signs and symptoms of severe pneumonia include severe tachypnea, SpO₂ <90 percent, grunting, and altered mental status.

A systematic review of studies predicting pneumonia severity in children suggests that hypoxemia, altered mental status, age <3 to 6 months, dyspnea, multilobar infiltrates, and moderate or large pleural effusions are the factors most predictive of severe pneumonia [34]. Nevertheless, standardized criteria for mild and severe pneumonia in children are lacking.

Severity of community-acquired pneumonia in infants and children	
Clinical features of mild pneumonia (all of the following)	Clinical features of severe pneumonia (1 or more of the following)
- Mild or absent respiratory distress: - No to mild tachypnea for age* - Mild or absent retractions - No grunting - No nasal flaring - No apnea - Mild shortness of breath	- Severe respiratory distress: - RR >70 breaths/minute for infants; RR >50 breaths/minute for older children - Severe suprasternal, intercostal, or subcostal retractions (>12 months) - Severe difficulty breathing (>12 months) - Grunting - Nasal flaring - Apnea - Significant shortness of breath
Normal color	Cyanosis
Normal mental status	Altered mental status
Normoxemia (oxygen saturation ≥90% in room air)	Hypoxemia (sustained oxygen saturation <90% in room air)
Normal feeding (infants); no vomiting	Not feeding (infants) or signs of dehydration (older children)
Normal heart rate	Tachycardia
Capillary refill <2 seconds	Capillary refill ≥2 seconds

Severity of community-acquired pneumonia in infants and children

Table 5 - larger image below

DIFFERENTIAL DIAGNOSIS

- **Upper respiratory tract infection with cough** – Many children with a viral upper respiratory tract infection have cough and fever; however, they do not have signs of respiratory distress. (See ["The common cold in children: Clinical features and diagnosis"](#), section on 'Symptoms and signs'.)
- **Bronchiolitis** – Infants and young children with bronchiolitis typically have cough and fever. The fever with bronchiolitis is usually $\leq 38.3^{\circ}\text{C}$ (101°F), which occurs early in the disease presentation. Wheezing is a prominent component of illness. Bronchiolitis is most common during the first year of life and rarely affects children who are older than two years. (See ["Bronchiolitis in infants and children: Clinical features and diagnosis"](#), section on 'Clinical features'.)
- **Asthma** – Most children with pneumonia do not wheeze, although wheezing can accompany atypical pneumonia. Children whose cough and wheezing are caused by asthma do not have a fever. (See ["Asthma in children younger than 12 years: Initial evaluation and diagnosis"](#), section on 'Symptoms'.)
- **Comorbid asthma and viral respiratory infection** – Young children with asthma who have viral respiratory infections can be misdiagnosed as having community-acquired pneumonia (CAP) [35]. Many such children have respiratory distress and may have hypoxemia. The diagnosis of CAP and treatment with antibiotics must be carefully considered in young children who have a prodrome compatible with a viral respiratory infection and are wheezing, even if there are pulmonary infiltrates (versus atelectasis) on chest radiograph. (See ["Asthma in children younger than 12 years: Initial evaluation and diagnosis"](#), section on 'Respiratory tract infections'.)
- **Foreign body aspiration** – Foreign body aspiration must be considered in young children with a cough. The aspiration event may not have been witnessed. (See ["Airway foreign bodies in children"](#), section on 'Presentation'.)
- **Other causes of tachypnea in infants and young children** – Other causes of tachypnea, with or without fever and cough, in infants and young children include [36]:
 - Heart failure (see ["Heart failure in children: Etiology, clinical manifestations, and diagnosis"](#), section on 'Clinical manifestations')
 - Sepsis (see ["Sepsis in children: Definitions, clinical manifestations, and diagnosis"](#) and ["Neonatal bacterial sepsis: Clinical features and diagnosis in neonates \$\geq 35\$ weeks of gestation"](#))
 - Metabolic acidosis (see ["Approach to the child with metabolic acidosis"](#), section on 'Acute metabolic acidosis')

- **Vaping-associated pulmonary illness** – In adolescents and young adults, use of electronic cigarette devices to inhale aerosolized substances (ie, "vaping") has been associated with a severe pulmonary illness that shares symptoms with pneumonia (eg, shortness of breath, fever, chest pain) [37,38]. Vaping-associated pulmonary illness is discussed separately. (See ["Vaping and e-cigarettes", section on 'Aerosol \(also known as vapor\) exposure'](#).)
- **Noninfectious mimics of pneumonia** – Noninfectious mimics of pneumonia can usually be distinguished from infectious pneumonia by history and/or associated clinical features. In some cases, laboratory studies or imaging may be necessary. The table presents a list of noninfectious conditions ( [table 7](#)).
- **Other considerations**
 - **Lemierre syndrome** – Lemierre syndrome (septic thrombophlebitis of the internal jugular vein) is an important consideration in adolescents and young adults whose illness began with pharyngitis. In Lemierre syndrome, the vessels of the carotid sheath become infected (typically with *Fusobacterium* spp), leading to bacteremia and metastatic spread of infection to the lungs and mediastinum. (See ["Lemierre syndrome: Septic thrombophlebitis of the internal jugular vein"](#).)
 - **Comorbid noninfectious lung disease** – Rare, noninfectious lung diseases may present with an intercurrent infectious illness. These conditions should be considered, especially if the acute illness is atypical or the radiographic and clinical findings do not resolve as expected with uncomplicated CAP.

Causes of noninfectious lung disease include:

- Pulmonary alveolar proteinosis (see ["Pulmonary alveolar proteinosis in children"](#))
- Eosinophilic pneumonia (see ["Idiopathic acute eosinophilic pneumonia"](#))
- Acute interstitial pneumonitis (see ["Acute interstitial pneumonia \(Hamman-Rich syndrome\)"](#))
- Cryptogenic organizing pneumonia (see ["Cryptogenic organizing pneumonia"](#))

COMPLICATIONS

Bacterial pneumonias are more likely than atypical bacterial or viral pneumonias to be associated with complications involving the respiratory tract.

- **Parapneumonic pleural effusion and empyema** – A child who presents with pneumonia may already have signs and symptoms of an effusion or empyema, or these may develop over the few days following initiation of treatment [39]. Hypoalbuminemia is common in children with parapneumonic effusions, and hypogammaglobulinemia may be encountered. The clinical features and evaluation of parapneumonic effusion ( [image 2A-B](#)) and empyema in children are discussed separately. (See ["Epidemiology, clinical presentation, and evaluation of](#)

parapneumonic effusion and empyema in children" and "Management and prognosis of parapneumonic effusion and empyema in children".)

- **Necrotizing pneumonia** – Necrotizing pneumonia, a serious complication of community-acquired pneumonia (CAP), should be considered in a child with prolonged fever or septic appearance [40]. Clinical manifestations of necrotizing pneumonia are similar to those of uncomplicated pneumonia, but they are more severe [41-43]. Necrotizing pneumonia usually follows pneumonia caused by particularly virulent bacteria [40] and is generally accompanied by parapneumonic effusion/empyema. *Streptococcus pneumoniae* (especially serotype 3 and serogroup 19) is the most common cause of necrotizing pneumonia ( image 3) [44-47]. *S. aureus* and group A *Streptococcus* may also cause necrotizing pneumonia, and necrotizing pneumonia due to *M. pneumoniae*, *Legionella*, and *Aspergillus* has been reported [43,46,48-51].

If the diagnosis remains uncertain, necrotizing pneumonia can be confirmed by Doppler ultrasonography, chest radiograph (which demonstrates a radiolucent lesion) ( image 1), or contrast-enhanced computed tomography (CT) ( image 3) [52,53]. The findings on chest radiograph may lag behind those of CT, and Doppler ultrasonography may detect necrotic changes earlier than CT [50,54].

- **Lung abscess** – Lung abscess should be suspected when consolidation is unusually persistent, when pneumonia remains persistently round or masslike, or when the volume of the involved lobe is increased (as suggested by a bulging fissure) [55,56]. Clinical manifestations of lung abscess are nonspecific and similar to those of pneumonia [55], and the course may be indolent. Manifestations include fever, cough, dyspnea, chest pain, anorexia, hemoptysis, and putrid breath [40,57-59], and they are often accompanied by a parapneumonic effusion [39,60,61]. The average duration of fever is four to eight days.

A lung abscess is composed of an accumulation of inflammatory cells accompanied by tissue destruction or necrosis that produces one or more cavities in the lung [55]. Abscess formation may result from inadequate or delayed treatment of lobar pneumonia or, more commonly, develops one to two weeks following an aspiration event. Other predisposing factors include airway obstruction and a congenitally abnormal lung [55]. Anaerobic flora of the upper respiratory tract and *S. aureus* are the organisms most frequently involved [36].

The diagnosis of lung abscess is suggested by a chest radiograph demonstrating a thick-walled cavity with an air-fluid level ( image 1) [55]. Contrast-enhanced CT confirms the diagnosis in questionable clinical situations or if confirmation of the diagnosis is necessary before invasive interventions are performed (eg, percutaneous aspiration, placement of a drainage catheter) [52]. Interventional radiology may be helpful in obtaining a specimen from the abscess cavity for diagnostic studies.

Eighty to 90 percent of lung abscesses in children resolve with antibiotic therapy alone and spontaneous drainage through the tracheobronchial tree, provided that any bronchial obstruction has resolved or, in the case of a foreign body, been removed [57].

- **Pneumatocele** – Pneumatoceles are thin-walled, air-containing cysts of the lung, which are most often discovered incidentally on imaging performed for evaluation of pneumonia or its complications. They are classically associated with *S. aureus* but may occur with a variety of organisms [62,63]. Pneumatoceles frequently occur in association with empyema [62]. Most pneumatoceles involute spontaneously [62,64,65]. However, on occasion, pneumatoceles may result in pneumothorax [63].
- **Hyponatremia** – Hyponatremia (serum sodium concentration ≤ 135 mEq/L) occurs in approximately 45 percent of children with CAP and one-third of children hospitalized with CAP, but is usually mild (ie, serum sodium concentration >130 and ≤ 135 mEq/L) [66-70]. Hyponatremia is associated with increased length of hospital stay, additional complications, and mortality and may be associated with inappropriate secretion of antidiuretic hormone [66,67,69,70]. (See "Pathophysiology and etiology of the syndrome of inappropriate antidiuresis (SIAD)", section on 'Pulmonary disease'.)

SETTING OF CARE

Most patients with mild pneumonia can be treated in the outpatient setting. Indications for hospitalization include moderate to severe respiratory distress, hypoxemia (peripheral arterial oxygen saturation [SpO₂] <90 percent on room air), toxic appearance, and inability of the caregiver to care for the child at home. Most children less than six months of age also require hospitalization. Details about these and additional indications for hospitalization are discussed separately (table 8). (See "Pneumonia in children >28 days old: Inpatient treatment", section on 'Indications for hospitalization'.)

Indications for inpatient management of community-acquired pneumonia in children	
Indication	Comment
Most children <3 to 6 months of age	Exception: a viral etiology or Chlamydia trachomatis is suspected and the child is relatively asymptomatic without toxemia
Inability to care for at home	Any child whose caregiver(s) cannot provide appropriate care or follow the management instructions
Hypoxia	SpO ₂ $<90\%$ on room air
Dehydration	Inability to maintain oral hydration or inability to feed an infant
Moderate to severe respiratory distress	Examples: RR >60 breaths per minute for infants <12 months of age, RR >100 breaths per minute for children ≥ 12 months of age, retractions, nasal flaring, difficult breathing, apnea, grunting
Toxic appearance	Toxic appearance is more common in bacterial pneumonia and may suggest more severe illness
Underlying conditions that predispose to a more severe course	Examples: cardiopulmonary disease, genetic syndromes, neurocognitive disorders
Underlying conditions that may be exacerbated by pneumonia	Example: metabolic disorder
Underlying conditions that may adversely affect response to treatment	Example: immunocompromised host
Complications of pneumonia present	Examples: effusions/empyema, abscessing process, lung abscess
Suspicion or confirmation of a pathogen that has increased virulence	Examples: <i>Staphylococcus aureus</i> , group A <i>Streptococcus</i>
Failure of outpatient therapy	Continued lack of response within 48 to 72 hours of antibiotic change

Indications for inpatient management of community-acquired pneumonia in children

Table 8 - larger image below

ADDITIONAL DIAGNOSTIC TESTING

Indications for additional diagnostic testing — Chest imaging is indicated for children admitted to the hospital to confirm the diagnosis, assess for complications, and document the size and character of parenchymal infiltrates. It is also indicated for children with a history of a previous pneumonia episodes to evaluate for a missed foreign body or anatomic abnormality (ie, risk factors for recurrent pneumonia).

A microbiologic evaluation to definitively determine the etiology of the pneumonia is indicated for children admitted to the hospital. Although the etiologic agent may be suggested by clinical presentation, epidemiologic considerations, and chest radiographic patterns, neither clinical nor

radiologic features reliably distinguish between typical bacterial, atypical bacterial, and viral pneumonia. (See '[Bacterial versus viral etiology](#)' above and '[Pneumonia in children: Epidemiology, pathogenesis, and etiology](#)', section on 'Etiologic agents'.)

Testing for critical microbes is indicated if there is a local outbreak. (See '[Critical microbes](#)' below.)

Chest imaging — Chest imaging typically consists of radiographs. Bedside lung ultrasonography has been used to detect lung consolidation in pediatric emergency and inpatient settings [71-75]. However, diagnostic accuracy appears to be affected by the level of experience of the sonographer [76].

- **Views** – Radiographic views depend upon the age of the child [77]. In children older than four years, the frontal posteroanterior (PA) upright chest view is usually obtained to minimize the size of the cardiac shadow [78]. In younger children, the anteroposterior (AP) supine view is preferred because the child's position does not affect the size of the cardiothoracic shadow in this age group and immobilization is easier, improving the likelihood of an adequate inspiration [78].

There is a lack of consensus regarding the need for lateral views to demonstrate infiltrates that may be obscured by the dome of the diaphragm or the cardiac shadow on AP or PA views [1,2,55,79]. In a review of chest radiographs in 201 children with pneumonia, the lateral film was abnormal in 91 percent of 109 children with definite pneumonia [80]. However, the AP or PA views also demonstrated pneumonia in all but three cases. A lateral view may be more valuable in settings where the radiographs are interpreted by nonradiologists.

Of note, the Pediatric Infectious Diseases Society and Infectious Diseases Society of America suggest PA and lateral views for all children who are hospitalized for management of community-acquired pneumonia (CAP) [1].

A lateral decubitus view (with the affected side down) may be needed to identify a pleural effusion. (See '[Epidemiology, clinical presentation, and evaluation of parapneumonic effusion and empyema in children](#)', section on 'Radiologic evaluation'.)

- **Findings** – Radiographic features do not reliably differentiate between typical bacterial, atypical bacterial, and viral pneumonia [8,81-83] and should be used in conjunction with clinical findings to make therapeutic decisions ([table 2](#)) [2,84-87].

In children with viral pneumonia, interstitial infiltrates are the primary radiographic finding.

Radiographic features that are more often associated with a bacterial etiology include:

- **Segmental and lobar consolidation** – Segmental and lobar consolidation are reasonably specific for bacterial pneumonia but lack sensitivity [86,88-90]. These findings are the most common ones associated with *M. pneumoniae* (37 percent) [91]. Radiologic features of

segmental consolidation are not always easy to distinguish from segmental collapse (atelectasis), which is seen in approximately 25 percent of children with bronchiolitis [92,93].

- **Pulmonary consolidation** – Pulmonary consolidation in young children sometimes appears to be spherical (ie, "round pneumonia") ([image 4](#)) [94,95]. Round pneumonias tend to be >3 cm, solitary, and posteriorly located [95,96]. *S. pneumoniae* is the most common bacterial etiology of round pneumonia; additional bacterial causes include other streptococci, *H. influenzae*, *S. aureus*, and *M. pneumoniae* [86,97].
- **Peribronchial infiltrates** – Peribronchial infiltrates are suggestive of *M. pneumoniae* pneumonia (27 percent) ([image 5](#)) [91]. However, *S. pneumoniae* has occasionally been associated with peribronchial infiltrates in children.
- **Hyperinflation with interstitial infiltrates** – Hyperinflation with interstitial infiltrates is characteristic of afebrile pneumonia of infancy typically caused by *C. trachomatis*. (See "[Chlamydia trachomatis infections in newborns and young infants](#)", section on 'Pneumonia'.)
- **Pneumatocoles, cavitations, large pleural effusions, and necrotizing processes** – Complications of pneumonia (eg, pneumatocoles, cavitations, large pleural effusions ([image 2A](#) and [image 2B](#)), necrotizing processes ([image 3](#))) suggest a typical bacterial etiology (eg, *S. pneumoniae*, *S. aureus*, *Streptococcus pyogenes*) and, rarely, *M. pneumoniae*. (See '[Complications](#)' above.)

Mediastinal and/or hilar adenopathy suggests a mycobacterial (eg, tuberculosis) or fungal (eg, histoplasmosis) etiology, although hilar adenopathy can also be seen in pneumococcal pneumonia.

- **Caveats** – Points to consider when interpreting radiographic findings include:

- Although a negative chest radiograph excludes pneumonia in most children ≥3 months of age [98,99], radiographic findings may lag behind clinical findings [100], and patients who are hypovolemic may have normal-appearing chest radiographs before volume repletion.

Mycoplasma pneumoniae pneumonia: Chest radiograph



Mycoplasma pneumoniae pneumonia: Chest radiograph

Image 5 - larger image below

Right-sided pneumonia with pleural effusion



Right-sided pneumonia with pleural effusion

Image 2a - larger image below

- There is variation in intraobserver and interobserver agreement of chest radiographic interpretation [2,101].
- Radiographic interpretation may be influenced by the clinical information that is provided to the radiologist [102,103].

Microbiologic testing — Our approach to microbiologic testing is generally consistent with that of the Pediatric Infectious Diseases Society and the Infectious Diseases Society of America clinical practice guidelines [1]. (See '[Society guideline links](#)' below.)

Blood cultures — We obtain blood cultures in children who require hospitalization because they are more likely to have bacteremia than those who are treated in the outpatient setting. (See "[Detection of bacteremia: Blood cultures and other diagnostic tests](#)".)

Observational studies conducted in the postpneumococcal conjugate vaccine era demonstrate prevalences of bacteremic pneumonia ranging from 2.2 to 7 percent in hospitalized children who had blood cultures obtained [104-106]. In a prospective study, blood cultures were obtained in 91 percent of 2358 children who were hospitalized with radiographically confirmed pneumonia and were positive in 2.2 percent (95% CI 1.6-2.9 percent) [106]. Most isolates were susceptible to penicillin. *S. pneumoniae* (46 percent), *S. aureus* (6 percent), and *S. pyogenes* (9 percent) were the most frequently isolated pathogens. In this and other observational studies, the prevalence of bacteremia was increased in patients with complicated pneumonia or pneumonia-related metastatic complications (eg, osteomyelitis) [104-110].

Targeting blood cultures to children who are at increased risk for bacteremia (eg, those with pleural effusion/empyema, admission to the intensive care unit, or immunosuppression) may be clinically effective and reduce costs [110,111]. However, determining which children are at risk for bacteremia at the time of admission can be difficult. In one multicenter study, 6 of the 26 patients with bacteremia were initially thought to have uncomplicated pneumonia [104].

Sputum Gram stain and culture — We obtain sputum samples for Gram stain and culture in children who require hospital admission if they are able to produce sputum [1]. Children younger than five years usually swallow sputum, so it is rarely available for examination.

Although good-quality sputum samples can be obtained by sputum induction [112,113], sputum induction is unpleasant. In addition, in a prospective study of the etiology of pneumonia in hospitalized children <18 years of age, bacterial pathogens were isolated from most induced sputum cultures regardless of sputum quality or whether the child had radiographically confirmed pneumonia, suggesting identification of oropharyngeal carriage rather than the pathogenic etiology of the pneumonia [113].

An aggressive approach to determining a definitive microbial diagnosis is necessary for immunocompromised patients with pneumonia that requires hospitalization. Invasive procedures may be required to obtain a satisfactory specimen. For patients with an

endotracheal tube in place, early flexible bronchoscopy for bronchoalveolar lavage with viral, fungal, and bacterial diagnostic studies is an option. However, the protected specimen brush technique has been utilized in some settings. Quantitative bacterial cultures may differentiate colonization from true lower respiratory tract infection. (See ["Flexible bronchoscopy in adults: Indications and contraindications"](#), section on 'Diagnostic indications' and ["Clinical presentation and diagnostic evaluation of ventilator-associated pneumonia"](#), section on 'Diagnostic evaluation'.)

Pleural fluid Gram stain and culture — Diagnostic (and possibly therapeutic) thoracentesis is generally warranted for children with more than minimal pleural effusion identified on radiographs. Specimens for culture should ideally be obtained before administration of antibiotics. If patients receive antibiotics before a pleural effusion is identified, molecular testing of the fluid for detection of *S. pneumoniae*, *S. pyogenes*, and *S. aureus* using polymerase chain reaction (PCR) techniques (when available) may be warranted. The evaluation of pleural fluid is discussed separately. (See ["Epidemiology, clinical presentation, and evaluation of parapneumonic effusion and empyema in children"](#), section on 'Pleural fluid analysis'.)

Rapid diagnostic testing — When available, we prefer rapid diagnostic tests to cultures because the results are available much sooner. Results of rapid diagnostic tests can be helpful in making treatment and/or cohorting decisions for infants and children who are admitted to the hospital with probable CAP, regardless of the suspected etiology [1,114]. Importantly, rapid viral testing cannot be used in isolation to exclude bacterial infection because patients, particularly those who are severely ill, may have a concomitant bacterial-viral infection [115].

Rapid diagnostic testing for respiratory viruses is usually done by real-time (RT) PCR. Multiplex RT-PCR viral panels can detect respiratory syncytial virus (RSV), influenza A and B, SARS-CoV-2, and other common respiratory viruses. These tests are performed on nasopharyngeal (NP) swabs, and results are typically available within a few hours. They can also be performed on pleural fluid. Immunofluorescence antigen tests are also available but are less sensitive than PCR.

Additionally, PCR tests are available for the detection of *S. pneumoniae* (performed on sputum, pleural fluid, and abscess contents) and *M. pneumoniae* (performed on a throat swab) [116]. In prospective studies, qualitative PCR of respiratory samples had a higher yield than culture for identification of *S. pneumoniae* [3,117,118]. However, this may merely represent respiratory tract colonization. *S. pneumoniae* PCR testing of pleural fluid or abscess contents, on the other hand, may help to differentiate colonization from infection [119-121].

M. pneumoniae PCR testing can be helpful in evaluating the younger child (ie, <4 years) with consistent symptoms given that *M. pneumoniae* is less common in children <4 years of age. PCR testing may also help to establish the diagnosis in patients with extrapulmonary manifestations of *M. pneumoniae*, particularly central nervous system manifestations [15]. (See ["Mycoplasma pneumoniae infection in children"](#), section on 'Clinical manifestations'.)

Additional details on rapid diagnostic tests for viral and atypical bacterial pathogens are discussed separately:

- RSV (see ["Respiratory syncytial virus infection: Clinical features and diagnosis in infants and children", section on 'Diagnosis'](#))
- Influenza viruses (see ["Seasonal influenza in children: Clinical features and diagnosis", section on 'Diagnosis'](#))
- SARS-CoV-2 (see ["COVID-19: Clinical manifestations and diagnosis in children", section on 'Approach to diagnosis'](#))
- Parainfluenza viruses (see ["Parainfluenza viruses in children", section on 'Diagnosis'](#))
- Adenovirus (see ["Diagnosis, treatment, and prevention of adenovirus infection", section on 'Differential diagnosis'](#))
- *M. pneumoniae* (see ["Mycoplasma pneumoniae infection in children", section on 'Diagnosis'](#))
- *Chlamydia* spp (see ["Pneumonia caused by *Chlamydia pneumoniae* in children", section on 'Diagnosis'](#))
- Human metapneumovirus (see ["Human metapneumovirus infections", section on 'Diagnosis'](#))

Less commonly used tests — Other tests that may be helpful in establishing less common microbiologic etiologies of CAP in children with appropriate clinical indications include:

- Interferon gamma release assay and [tuberculin skin test](#) – If pulmonary tuberculosis is a consideration, a serum interferon gamma release assay or a tuberculin skin test is warranted. Indications include exposure to an individual with active tuberculosis ([table 9](#)), being foreign born, having a foreign-born caregiver, a history of foreign travel, and exposure to an individual who traveled to an endemic area. Additional diagnostic testing for tuberculosis in children is discussed separately. (See ["Tuberculosis disease \(active tuberculosis\) in children: Epidemiology, clinical manifestations, and diagnosis".](#))
- Urine antigen testing for legionellosis due to serogroup 1 – If *Legionella* is a consideration (eg, pneumonia in an immunocompromised host; known/suspected outbreak in a building, cruise ship, etc), urine antigen testing for legionellosis due to serogroup 1 is recommended. Diagnosis of *Legionella* infection is discussed in detail separately. (See ["Clinical manifestations and diagnosis of *Legionella* infection", section on 'Approach to testing'](#) and ["Microbiology, epidemiology, and pathogenesis of *Legionella* infection", section on 'Pathogen'.](#))
- Serum and urine antigen and antibody testing for histoplasmosis – If histoplasmosis is a consideration, serum and urine antigen and antibody testing should be conducted (ie,

complement fixation, immunodiffusion, and enzyme immunoassay [EIA] immunoglobulin M [IgM]/immunoglobulin G [IgG]). Indications for testing include exposure to soil contaminated with bird droppings or bat guano in an endemic area [122]. Diagnostic testing for histoplasmosis is discussed in more detail separately. (See "[Pathogenesis and clinical features of pulmonary histoplasmosis](#)" and "[Diagnosis and treatment of pulmonary histoplasmosis](#)", section on 'Antigen detection'.)

Urine antigen testing for *S. pneumoniae* in children should **not be** performed because of concern for false-positive reactions, some of which may merely indicate colonization with *S. pneumoniae* [1,2]; additional studies are needed to evaluate the potential usefulness of this diagnostic technique in children with and without CAP [123].

We do not suggest routine serologic testing for specific pathogens (eg, *S. pneumoniae*, *M. pneumoniae*, *C. pneumoniae*) because the results usually do not influence management [124,125]. *S. pneumoniae* has too many potential infecting serotypes to make antibody determinations practical. IgM antibody testing for *M. pneumoniae* may give false-positive results. Serologic tests for *Chlamydia* spp are not readily available. Serologic diagnosis of viral pathogens is not practical because acute and convalescent specimens are needed.

We do not obtain NP cultures for etiologic diagnosis in children with pneumonia. Bacterial organisms cultured from the nasopharynx may be normal upper respiratory flora. Although the results of NP cultures for viruses and atypical bacteria may be helpful, they may not be available soon enough to assist with management decisions.

The addition of cell-free plasma sequencing and transcriptional profile analysis to diagnostic capabilities awaits the results of further studies in children with CAP and more widespread availability [126,127]. These laboratory approaches may be most useful in immunocompromised hosts with CAP for whom the risks of invasive procedures may outweigh the benefits [128].

Critical microbes — Diagnostic testing for an unusual pathogen (critical microbe) should be conducted if the patient has characteristic symptoms, a close contact, or if there is an outbreak. These pathogens require treatment that differs from standard empiric regimens for CAP and have public health implications.

Diagnostic testing for the following critical microbes is discussed separately:

- Measles ( [picture 1A-B](#)) (see "[Measles: Clinical manifestations, diagnosis, treatment, and prevention](#)")
- Varicella ( [picture 2](#)) (see "[Clinical features of varicella-zoster virus infection: Chickenpox](#)" and "[Diagnosis of varicella-zoster virus infection](#)")
- Fungal etiologies (*Coccidioides immitis*, *Blastomyces dermatitidis*, *Histoplasma capsulatum*) (see "[Primary pulmonary coccidioidal infection](#)" and "[Mycology, pathogenesis, and epidemiology](#)")

of blastomycosis" and "Treatment of blastomycosis" and "Diagnosis and treatment of pulmonary histoplasmosis")

- Avian influenza (see "Avian influenza: Clinical manifestations and diagnosis")
- Hantavirus (see "Hantavirus cardiopulmonary syndrome")
- Agents of bioterrorism (see "Detection of and response to biologic terrorism")

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "Society guideline links: Pediatric pneumonia".)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or email these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient education" and the keyword[s] of interest.)

- Basics topic (see "Patient education: Pneumonia in children (The Basics)")

SUMMARY AND RECOMMENDATIONS

• Clinical presentation

- **Clinical features** – Children with community-acquired pneumonia (CAP) typically have a combination of fever and cough and possibly other respiratory findings (eg, tachypnea, increased work of breathing), depending on the severity of the infection. Additional signs and symptoms vary by age. (See 'Clinical features' above.)
- **Etiology (bacterial versus viral)** – Clinical features that are traditionally considered characteristic of typical bacterial pneumonia, atypical bacterial pneumonia, and viral pneumonia frequently overlap and cannot be used reliably to distinguish between the

various etiologies. As many as 50 percent of infections are mixed bacterial-viral infections. However, features that are typically seen with each type of pneumonia can help to make a clinical diagnosis ([table 2](#)). (See '[Bacterial versus viral etiology](#)' above.)

- **Clinical evaluation**

- **Objectives** – The clinical evaluation should focus on features that help to define the clinical syndrome (eg, pneumonia, bronchiolitis) and determine the likely etiology (eg, bacteria, virus) ([table 2](#)). (See '[Objectives](#)' above and "[Pneumonia in children: Epidemiology, pathogenesis, and etiology](#)", section on '[Etiologic agents](#)'.)
- **History** – Historical features help determine the presumptive microbiologic etiology ([table 3](#) and [table 4](#)), the likelihood of infection with an organism that is resistant to antibiotics, and the risk of a missed foreign body or anatomical abnormality. (See '[History](#)' above.)
- **Physical examination** – Important aspects of the physical examination include general appearance (eg, state of awareness), temperature (eg, afebrile, fever $\geq 39^{\circ}\text{C}$), degree of respiratory distress, and lung examination ([table 6](#)). (See '[Physical examination](#)' above.)
- **Radiologic evaluation** – Radiographs are not necessary for children with mild pneumonia who are well enough to be treated as outpatients unless there is a history of recurrent pneumonia. Recurrent pneumonia may indicate a missed foreign body or anatomical abnormality. (See '[Radiographs](#)' above.)
- **Establishing the diagnosis** – The diagnosis of pneumonia is established in a child who has fever and cough, evidence by history or on physical examination of an infectious pulmonary process ([table 3](#) and [table 6](#)), and symptoms or signs of respiratory distress. (See '[Establishing the diagnosis](#)' above.)

Although no single symptom or sign is pathognomonic for pneumonia in children, the longer that fever, cough, and respiratory findings are present, the greater the likelihood of pneumonia.

- **Assessing severity** – The history and physical examination are used to assess the severity of illness ([table 5](#)), which determines, in part, whether hospitalization is indicated. (See '[Assessing severity](#)' above.)
- **Differential diagnosis** – The differential diagnosis includes common respiratory illnesses (eg, upper respiratory tract infection with cough, bronchiolitis, asthma, comorbid asthma with viral respiratory infection) and less common diagnoses (eg, foreign body aspiration, vaping-associated pulmonary illness, other causes of tachypnea, and noninfectious mimics of pneumonia) ([table 7](#)). (See '[Differential diagnosis](#)' above.)

- **Complications** – Complications of CAP in children include pleural effusion and empyema, necrotizing pneumonia, lung abscess, pneumatocele, and hyponatremia. (See '[Complications](#)' above.)
- **Indications for additional diagnostic testing** – Most children with mild pneumonia who are well enough to be treated as outpatients do not require additional diagnostic testing. Children with pneumonia who are admitted to the hospital should have chest imaging and a microbiologic evaluation. (See '[Additional diagnostic testing](#)' above.)

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Topic 5986 Version 64.0

GRAPHICS

Table 1: Pneumonia terminology

Term	Definition
Classification by site of acquisition	
Community-acquired pneumonia (CAP)	An acute infection of the pulmonary parenchyma acquired outside of health care settings
Nosocomial pneumonia	An acute infection of the pulmonary parenchyma acquired in hospital settings, which encompasses hospital-acquired pneumonia and ventilator-associated pneumonia
Hospital-acquired pneumonia (HAP)	Pneumonia acquired ≥ 48 hours after hospital admission; includes both HAP and VAP
Ventilator-associated pneumonia (VAP)	Pneumonia acquired ≥ 48 hours after endotracheal intubation
Non-ventilated hospital-acquired pneumonia (nvHAP)	Pneumonia that develops in hospitalized patients who are not on mechanical ventilation nor underwent extubation within 48 hours before pneumonia developed
Health care-associated pneumonia (HCAP)	Retired term, which referred to pneumonia acquired in health care facilities (eg, nursing homes, hemodialysis centers) or after recent hospitalization*
Classification by etiology	
Atypical pneumonia	Pneumonia caused by "atypical" [¶] pathogens including <i>Legionella</i> spp, <i>Mycoplasma pneumoniae</i> , <i>Chlamydia pneumoniae</i> , <i>Chlamydia psittaci</i> , <i>Coxiella burnetii</i>
Aspiration pneumonia	Pneumonia resulting from entry of gastric or oropharyngeal fluid, which may contain bacteria and/or exogenous substances (eg, ingested food particles or liquids, mineral oil, salt or fresh water) or be of low pH, into the lower airways
Chemical pneumonitis	Aspiration of substances (eg, acidic gastric fluid) that cause an inflammatory reaction in the lower airways, independent of bacterial infection
Bacterial aspiration pneumonia	An active infection caused by inoculation of large amounts of bacteria into the lungs via orogastric contents

* The term HCAP was used to identify patients at risk for infection with multidrug-resistant pathogens. This categorization may have been overly sensitive, leading to increased, inappropriately broad antibiotic use.

¶ The origin of the term "atypical" is a matter of debate. The term may refer to the fact that these organisms are not "typical" bacteria (eg, *Streptococcus pneumoniae*), which can be identified by standard microbiologic techniques. Others suggest that atypical refers to the mild nature of the pneumonia caused by some of these organisms compared with pneumonia caused by *S. pneumoniae*.

Graphic 130821 Version 6.0

Table 2: Clinical and radiographic clues to the etiology of pneumonia in children

Etiology	Clinical features	Radiographic features
Bacterial (most commonly <i>Streptococcus pneumoniae</i>)	- Children of all ages - Abrupt onset - Ill appearance - Chills - Moderate to severe respiratory distress - Focal auscultatory findings - Localized chest pain - WBC count >15,000/mL (if obtained) - Elevated acute phase reactants (if obtained)	- Alveolar infiltrates - Segmental consolidation - Lobar consolidation "Round" pneumonia Complications: - Pleural effusion/empyema - Lung abscess - Necrotizing pneumonia - Pneumatocele
Atypical bacterial (most commonly <i>Mycoplasma pneumoniae</i>)	- Children of all ages (most common in children >5 years) - Abrupt onset with gradually worsening constitutional findings (malaise, myalgia, headache, rash, conjunctivitis, photophobia, sore throat) and nonproductive cough - Wheezing - Extrapulmonary manifestations or complications (eg, polymorphous mucocutaneous eruptions, hemolytic anemia, hepatitis, pancreatitis, myopericarditis, aseptic meningitis)	- Lobar or segmental consolidation (37%) - Perihilar or peribronchial infiltrates (27%) - Localized reticulonodular infiltrates (21%) - Patchy infiltrates (15%)
Viral	- Usually children <5 years - Gradual onset - Preceding upper airway symptoms - Nontoxic appearance - Diffuse, bilateral auscultatory findings - Wheezing - May have associated rash (eg, measles, varicella)	- Interstitial infiltrates - Associated bronchiolitis: - Patchy atelectasis - Peribronchial infiltrations with air bronchograms - Hyperinflation with flattening of the diaphragms
Afebrile pneumonia of infancy (most commonly <i>Chlamydia trachomatis</i>)	- Usually in infants 1 to 3 months of age - Insidious onset - Tachypnea, diffuse crackles - Rhinorrhea - Staccato cough pattern - Peripheral eosinophilia (if CBC obtained)	Hyperinflation with interstitial infiltrates

Fungal	▪ Appropriate geographic or environmental exposure	▪ Mediastinal or hilar adenopathy
<i>Mycobacterium tuberculosis</i>	▪ Children of any age ▪ Chronic cough ▪ Constitutional symptoms ▪ Exposure history	▪ Mediastinal or hilar adenopathy

CBC: complete blood count; WBC: white blood cell.

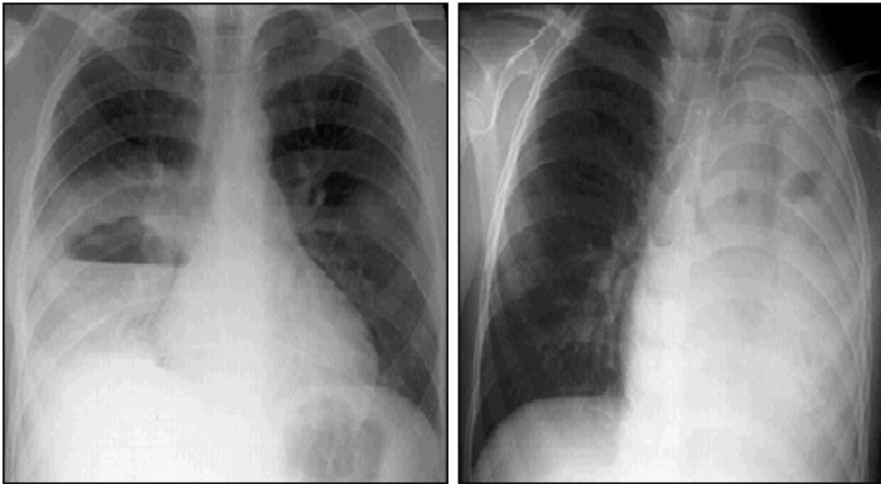
* The clinical features frequently overlap and cannot reliably distinguish between bacterial, atypical bacterial, and viral etiologies; up to one-half of community-acquired pneumonias in children may be mixed bacterial/viral infections. Chest radiography generally is not helpful in determining the potential causative agent of pneumonia. Nonetheless, these features may facilitate decisions regarding empiric therapy.

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Graphic 52021 Version 18.0

Image 1: Pneumococcal pneumonia: Complications



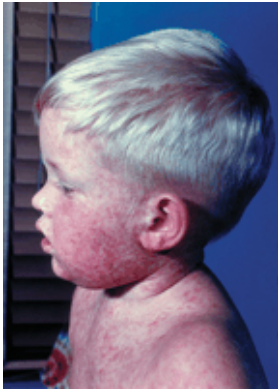
Radiographic images of the complications of pneumococcal pneumonia.

(Left panel) Lung abscess with an air-fluid level in the right lung. Abscess cavity material is nearly always culture positive, and patients commonly defervesce within 48 hours of interventional drainage.

(Right panel) Radiograph of necrotizing pneumonia in the left lung.

Graphic 53664 Version 9.0

Picture 1A: Boy with measles



Source: Centers for Disease Control and Prevention.

Graphic 57803 Version 3.0

Picture 1B: Measles exanthem



Blanching erythematous macules with some confluent areas on the trunk in a patient with measles.

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Graphic 57093 Version 4.0

Picture 2: Primary varicella lesions



Vesicular lesions on an erythematous base are characteristic of chickenpox. The lesions occur in crops and are present in a variety of stages from maculopapular to vesicular or even pustular. Central necrosis and early crusting is also visible.

Courtesy of Lee T Nesbitt, Jr. The Skin and Infection: A Color Atlas and Text, Sanders CV, Nesbitt LT Jr (Eds), Williams & Wilkins, Baltimore 1995.

<http://www.lww.com>

Graphic 55533 Version 5.0

Table 3: Important aspects of the history in a child with pneumonia

Historical feature	Possible significance
Age of the child	Viral etiologies are most common in infants and preschool children Atypical bacterial pathogens are more common in school-age children
Recent viral upper respiratory tract infection	May predispose to bacterial superinfection with <i>Streptococcus pneumoniae</i> or <i>Staphylococcus aureus</i>
Associated symptoms	<i>Mycoplasma pneumoniae</i> is often associated with extrapulmonary manifestations (eg, headache, sore throat, conjunctivitis, photophobia, rash)
Cough, chest pain, shortness of breath, difficulty breathing	"Classic" features of pneumonia, but nonspecific
Increased work of breathing in the absence of stridor or wheezing	Suggestive of more severe pneumonia
Fluid and nutrition intake	Difficulty or inability to feed suggests severe illness
Choking episode	May indicate foreign body aspiration
Duration of symptoms	Chronic cough (>4 weeks) suggests etiology other than acute pneumonia (refer to UpToDate topic on causes of chronic cough in children)
Previous episodes	Recurrent episodes may indicate aspiration, congenital or acquired anatomic abnormality, cystic fibrosis, immunodeficiency, asthma, missed foreign body
Immunization status	Completion of the primary series of immunizations for <i>Haemophilus influenzae</i> type b, <i>S. pneumoniae</i> , <i>Bordetella pertussis</i> , and seasonal influenza decreases, but does not eliminate, the risk of infection with these organisms
Previous antibiotic therapy	Increases the likelihood of antibiotic-resistant bacteria
Maternal history of chlamydia during pregnancy (for infants <4 months of age)	May indicate <i>Chlamydia trachomatis</i> infection
Foreign travel or known exposure to tuberculosis	May indicate <i>Mycobacterium tuberculosis</i> infection
Ill contacts	More common with viral etiologies
Travel to or residence in certain areas that suggest endemic pathogens	- Measles: Resource-limited countries - Coccidioidomycosis: Southwestern United States, northern Mexico, Central and South America - Blastomycosis: Southeastern and Central United States; states bordering the Great Lakes - Histoplasmosis: Ohio, Missouri, and Mississippi River valleys in the United States; Canada; Central America; eastern and southern Europe; parts of Africa; eastern Asia; and Australia

	▪ Hantavirus: West of the Mississippi River; four corners region of United States (where borders of Colorado, New Mexico, Arizona, and Utah meet) ▪ Middle East respiratory syndrome (MERS): Countries in or near Arabian Peninsula
Animal exposure	May indicate histoplasmosis, psittacosis, Q fever
Day care center attendance	Exposure to viruses and antibiotic-resistant bacteria

Graphic 52510 Version 15.0

Table 4: Etiologic agents of pediatric pneumonia

Microbial agent	Susceptible hosts
Bacteria	
<i>Chlamydia trachomatis</i>	First 3 months after birth
<i>Mycoplasma hominis</i>	First 3 months after birth
<i>Treponema pallidum</i>	First 3 months after birth
<i>Ureaplasma urealyticum</i>	First 3 months after birth
<i>Haemophilus influenzae</i> type b	Unimmunized children <4 years
<i>Staphylococcus aureus</i>	Primarily children <5 years
<i>Streptococcus pyogenes</i>	Primarily children <5 years
<i>Chlamydia pneumoniae</i>	Primarily children ≥5 years
<i>Mycoplasma pneumoniae</i>	Primarily children ≥5 years
<i>Streptococcus pneumoniae</i>	All
<i>Mycobacterium tuberculosis</i>	All
Viruses	
Respiratory syncytial virus	Primarily children <5 years
Influenza A and B	Primarily children <5 years
Human metapneumovirus	Primarily children <5 years
Parainfluenza viruses 1, 2, and 3	Primarily children <5 years
Coronaviruses (229E C43, NL63, HKU1, SARS-CoV-2)	Primarily children <5 years (SARS-CoV-2: all children)
Adenoviruses	Primarily children <5 years
Rhinovirus	Primarily children <5 years
Cytomegalovirus	Primarily children <1 year
Fungi	
Endemic mycoses (eg, histoplasmosis, coccidioidomycosis, blastomycosis)	All

SARS-CoV-2: severe acute respiratory syndrome coronavirus 2.

* Excluding neonatal pneumonia (in infants <28 days of age).

Data from:

1. Byington CL, Bradley JS. Pediatric community-acquired pneumonia. In: Feigin and Cherry's Textbook of Pediatric Infectious Diseases, 7th ed, Cherry JD, Harrison GJ, Kaplan SL, et al. (Eds), Elsevier Saunders, Philadelphia 2014. p.283.
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Graphic 54228 Version 18.0

Table 5: Severity of community-acquired pneumonia in infants and children

Clinical features of mild pneumonia (all of the following)	Clinical features of severe pneumonia (1 or more of the following)
Mild or absent respiratory distress: - No to mild tachypnea for age* - Mild or absent retractions - No grunting - No nasal flaring - No apnea - Mild shortness of breath	Severe respiratory distress: - RR >70 breaths/minute for infants; RR >50 breaths/minute for older children - Severe suprasternal, intercostal, or subcostal retractions (<12 months) - Severe difficulty breathing (≥12 months) - Grunting - Nasal flaring - Apnea - Significant shortness of breath
Normal color	Cyanosis
Normal mental status	Altered mental status
Normoxemia (oxygen saturation ≥90% in room air)	Hypoxemia (sustained oxygen saturation <90% in room air)
Normal feeding (infants); no vomiting	Not feeding (infants) or signs of dehydration (older children)
Normal heart rate	Tachycardia
Capillary refill <2 seconds	Capillary refill ≥2 seconds

RR: respiratory rate.

* World Health Organization definitions of tachypnea according to age are as follows: <2 months: >60 breaths/minute; 2 to 12 months: >50 breaths/minute; 1 to 5 years: >40 breaths/minute; >5 years: >20 breaths/minute.^[1]

References:

1. World Health Organization. *The management of acute respiratory infections in children: Practical guidelines for outpatient care*. World Health Organization 1995.

Adapted from:

- Bradley JS, Byington CL, Shah SS, et al. *The management of community-acquired pneumonia in infants and children older than 3 months of age: Clinical practice guidelines by the Pediatric Infectious Diseases Society and the Infectious Diseases Society of America*. *Clin Infect Dis* 2011; 53:e25.
- Harris M, Clark J, Coote N, et al. *British Thoracic Society guidelines for the management of community acquired pneumonia in children: Update 2011*. *Thorax* 2011; 66:ii1.

Graphic 72015 Version 8.0

Table 6: Important aspects of the physical examination in a child with suspected pneumonia

Physical examination feature	Possible significance
General appearance (state of awareness, cyanosis)*	Most children with radiographically confirmed pneumonia appear ill
Vital signs	
Temperature	High fever ($\geq 39^{\circ}\text{C}$ [102.2°F]) may be the only sign of pneumonia in young children; however, it is variably present and nonspecific
Respiratory rate	Tachypnea¶ less predictive of radiographically confirmed pneumonia than hypoxemia or increased work of breathing Tachypnea of >70 breaths/minute correlates with hypoxemia in infants <1 year of age Absence of tachypnea helps to exclude pneumonia
Degree of respiratory distress	Respiratory distress is more specific than fever or cough for lower respiratory infection
Tachypnea	Described in section on "Vital signs"
Hypoxemia	Predictive of pneumonia
Increased work of breathing:	
Retractions	More common in children with pneumonia than without; absence does not exclude pneumonia
Nasal flaring	More common in children with pneumonia than without; absence does not exclude pneumonia
Grunting	Sign of severe disease and impending respiratory failure
Accessory muscle use	Sign of severe disease
Head bobbing	Sign of severe disease
Lung examination	
Cough	Nonspecific finding of pneumonia
Auscultation	Findings suggestive of pneumonia include: crackles (rales, crepitations), decreased breath sounds, bronchial breath sounds, egophony, bronchophony, and whispered pectoriloquy Wheezing more common in viral and atypical pneumonias
Tactile fremitus	Suggestive of parenchymal consolidation
Dullness to percussion	Suggestive of parenchymal consolidation or pleural effusion
Mental status	Altered mental status may be a sign of hypoxia

* For young infants: Ability to attend to the environment, feed, vocalize, and be consoled.

¶ World Health Organization definitions of tachypnea according to age are as follows: <2 months: >60 breaths/min; 2 to 12 months: >50 breaths/min; 1 to 5 years: >40 breaths/min; ≥5 years: >20 breaths/min.

Graphic 65313 Version 7.0

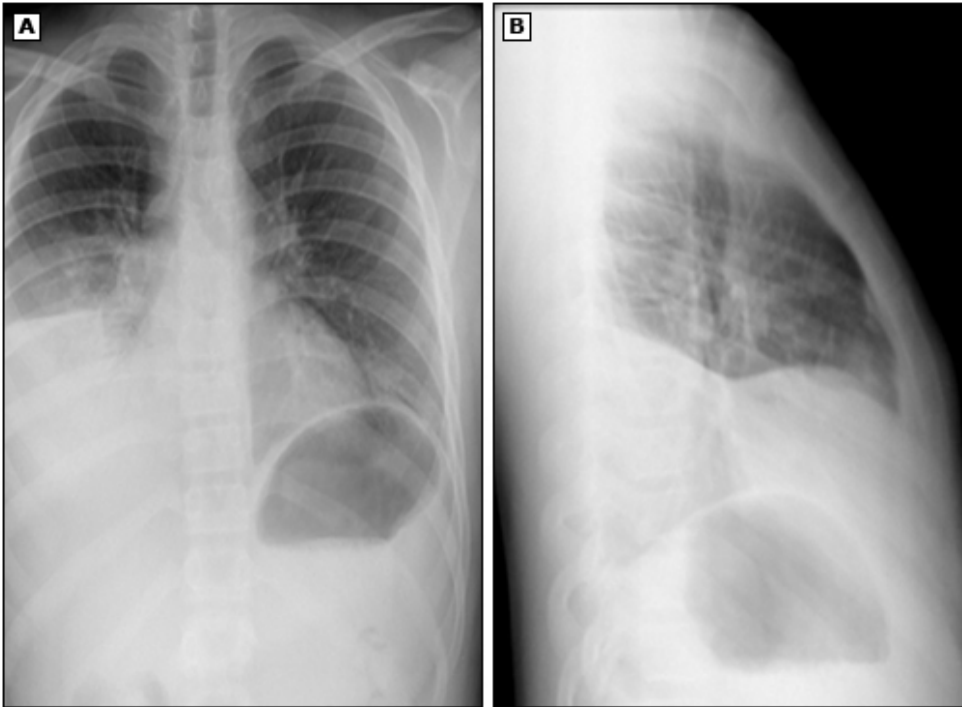
Table 7: Noninfectious conditions that can mimic pneumonia in children

Anatomical considerations	Drugs and chemical exposures
Prominent thymus	Nitrofurantoin
Breast shadows	Bleomycin
Bronchogenic cyst	Cytotoxic drugs
Vascular ring	Opiates
Pulmonary sequestration	Radiation therapy
Congenital lobar emphysema	Smoke inhalation
Atelectasis (due to a foreign body or mucus plug)	"Vaping" (ie, use of electronic cigarette devices to inhale aerosolized substances)
Aspiration of gastric contents	Lipoid pneumonia
Gastroesophageal reflux	Vasculitic disorders
Tracheoesophageal fistula	Systemic lupus erythematosus
Cleft palate	Granulomatosis with polyangiitis (Wegener)
Neuromuscular disorders	Juvenile idiopathic arthritis
Chronic pulmonary disorders	Others
Asthma	Hypersensitivity pneumonitis
Bronchiectasis	Neoplasm
Bronchopulmonary dysplasia	Pulmonary edema due to heart failure
Cystic fibrosis	Pulmonary infarction
Pulmonary fibrosis	Acute respiratory distress syndrome
Alpha-1 antitrypsin deficiency	Graft-versus-host disease
Pulmonary hemosiderosis	Poor inspiratory film
Alveolar proteinosis	Near drowning event
Desquamative interstitial pneumonitis	Underpenetrated film
Sarcoidosis	
Histiocytosis X	

Data from:

1. Klein JO. Bacterial pneumonias. In: *Textbook of Pediatric Infectious Diseases*, 5th ed, Feigin RD, Cherry JD, Demmler GJ, Kaplan SL (Eds), WB Saunders, Philadelphia 2004. p.299.
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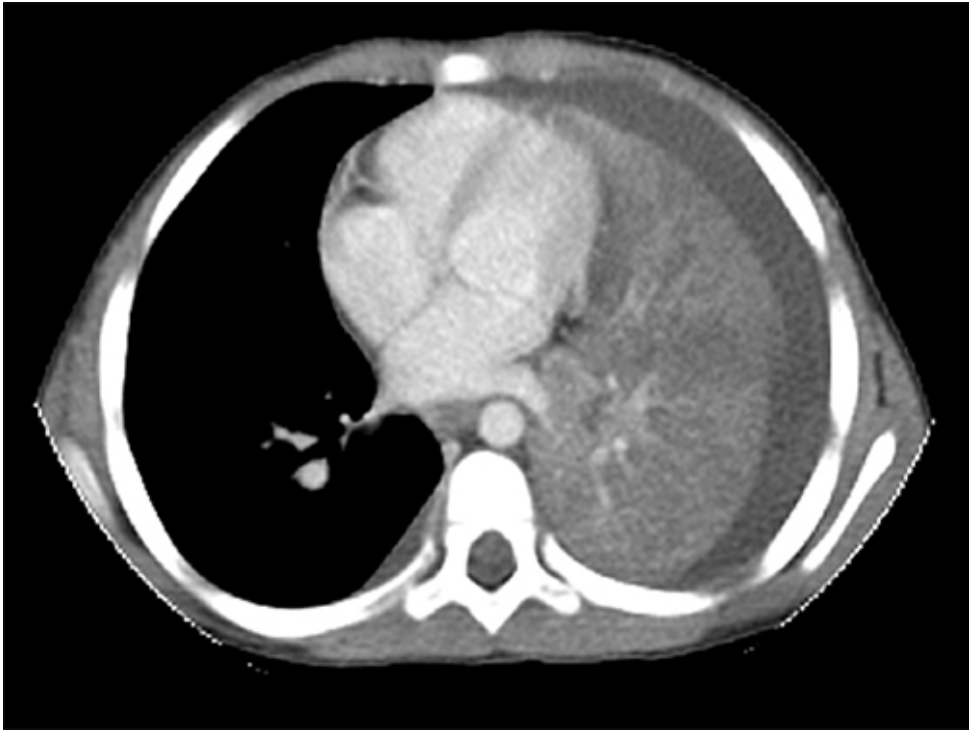
Image 2A: Right-sided pneumonia with pleural effusion



Courtesy of Dwight A Powell, MD.

Graphic 86360 Version 1.0

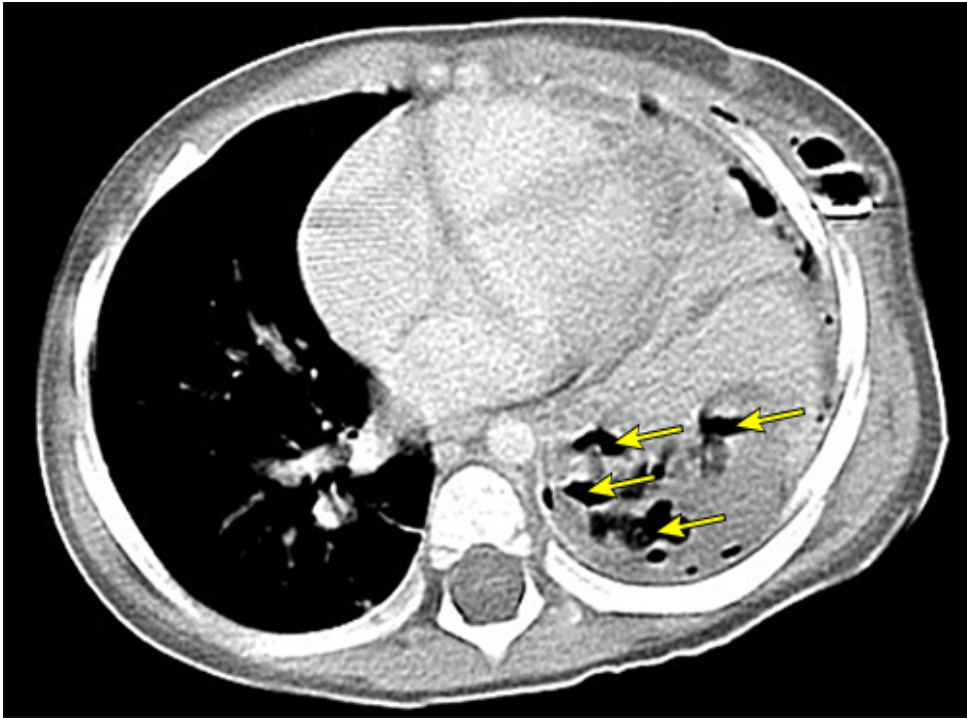
Image 2B: Computed tomography: Left-sided pneumonia with pleural effusion



Courtesy of Dwight A Powell, MD.

Graphic 86361 Version 1.0

Image 3: Computed tomography: Left- *Streptococcus pneumoniae* necrotizing pneumonia sided



Computed tomography of the chest demonstrating left-sided necrotizing pneumonia. Note the consolidation of the left lung parenchyma with focal areas of liquefaction (arrows).

Courtesy William J Barson, MD.

Graphic 86362 Version 3.0

Table 8: Indications for inpatient management of community-acquired pneumonia in children

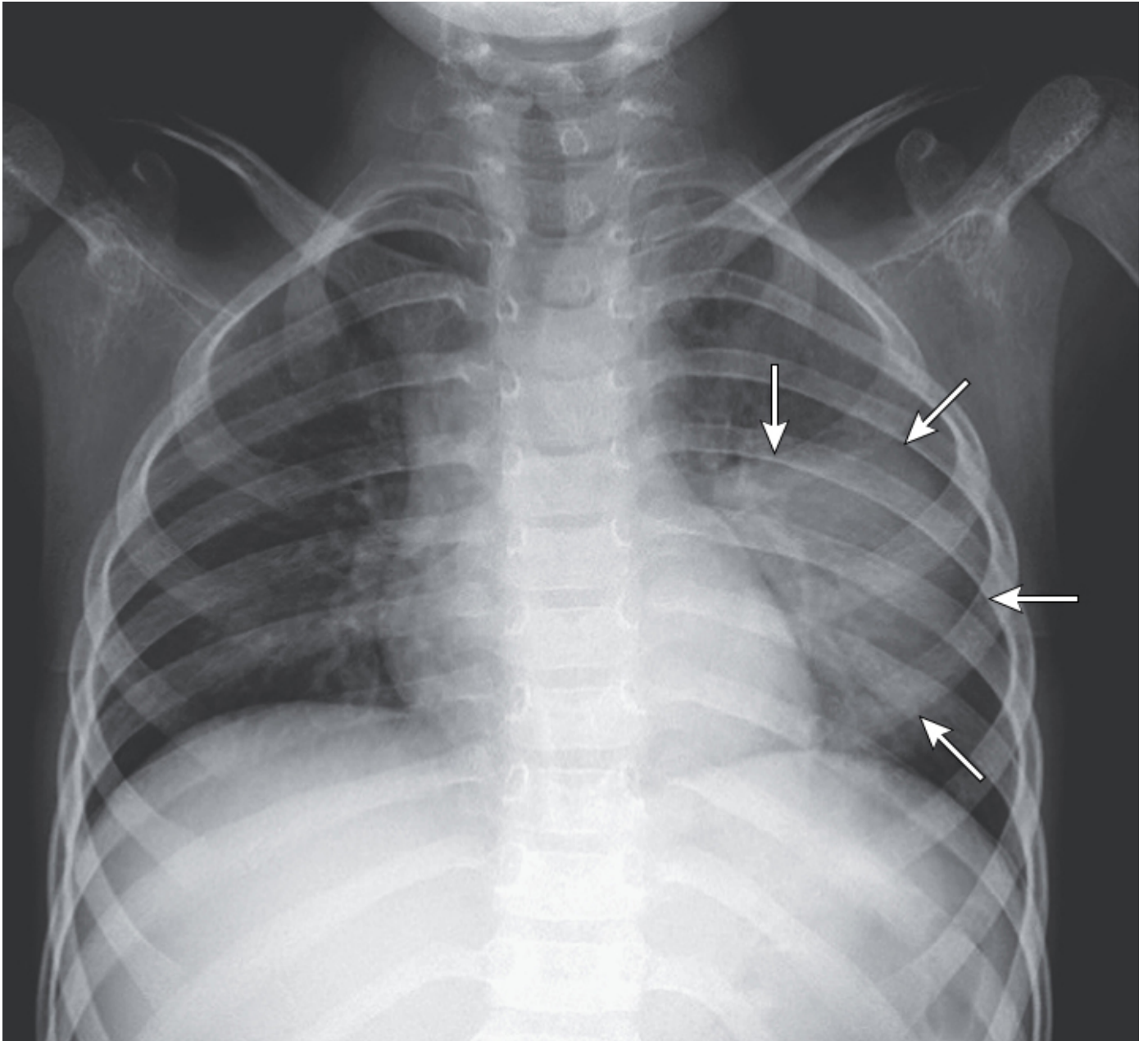
Indication	Comment
Most children <3 to 6 months of age	Exception: a viral etiology or <i>Chlamydia trachomatis</i> is suspected and the child is relatively asymptomatic without hypoxia
Inability to care for at home	Any child whose caregiver(s) cannot provide appropriate care or follow the management instructions
Hypoxia	SpO ₂ <90% on room air
Dehydration	Inability to maintain oral hydration or inability to feed an infant
Moderate to severe respiratory distress	Examples: RR >70 breaths per minute for infants <12 months of age, RR >50 breaths per minute for children ≥12 months of age, retractions, nasal flaring, difficulty breathing, apnea, grunting
Toxic appearance	Toxic appearance is more common in bacterial pneumonia and may suggest more severe illness
Underlying conditions that predispose to a more severe course	Examples: cardiopulmonary disease, genetic syndromes, neurocognitive disorders
Underlying conditions that may be exacerbated by pneumonia	Example: metabolic disorder
Underlying conditions that may adversely affect response to treatment	Example: immunocompromised host
Complications of pneumonia present	Examples: effusion/empyema, necrotizing process, lung abscess
Suspicion or confirmation of a pathogen that has increased virulence	Examples: <i>Staphylococcus aureus</i> , group A <i>Streptococcus</i>
Failure of outpatient therapy	Continued lack of response within 48 to 72 hours of antibiotic change

We recommend that children with any of the above indications be hospitalized for management of CAP.

CAP: community-acquired pneumonia; RR: respiratory rate; SpO₂: peripheral capillary oxygen saturation.

Graphic 147343 Version 1.0

Image 4: Round pneumonia in a 4-year-old child

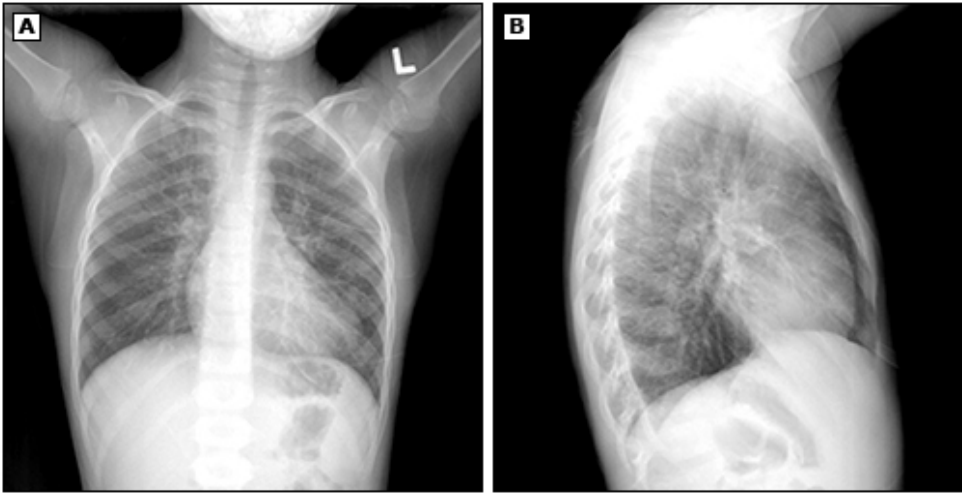


There is a round opacity in the left lower lobe with subtle air bronchograms. The arrows indicate the borders of the opacity.

From: Camacho-Cruz J, Guerrero-Sanchez D. Round pneumonia. N Engl J Med 2025; 392:e39. Copyright © 2025 Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.

Graphic 149220 Version 1.0

Image 5: *Mycoplasma pneumoniae* pneumonia: Chest radiograph



Diffuse bilateral interstitial infiltrates with *M. pneumoniae* infection.

Courtesy of Dwight A Powell, MD.

Graphic 86363 Version 4.0

Table 9: Tuberculosis terminology

Range of terms used in the literature	Clinical definition	Typical laboratory data
- Latent tuberculosis infection (LTBI) - Tuberculosis infection (TBI)	- Absence of symptoms or signs of illness - Evidence of specific cell-mediated immunologic response following exposure to <i>Mycobacterium tuberculosis</i>-derived protein antigens in solution	Positive IGRA or TST
- Active tuberculosis (TB) - Active tuberculosis disease - Active tuberculosis infection - Tuberculosis disease	Presence of symptoms or signs of illness due to <i>M. tuberculosis</i>	Detection of <i>M. tuberculosis</i> in clinical specimens via microscopy, culture, or molecular testing

Tuberculosis terminology is inconsistent in the literature^[1].

IGRA: interferon-gamma release assay; TST: tuberculin skin test.

Reference:

- Behr MA, Kaufmann E, Duffin J, et al. Latent tuberculosis: Two centuries of confusion. *Am J Respir Crit Care Med* 2021; 204:142.

Graphic 132526 Version 4.0

Contributor Disclosures

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